

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



**“IoT Enabled Dynamic Wireless Charging of Electric Vehicle
Using Solar Integration”**

*Submitted in partial fulfillment for the award of the degree
Bachelor of Engineering*

In

*Electronics and Communication Engineering
submitted by*

Bhavani V R

1AT22EC 013

Kavana T R

1AT22EC 039

Mahima Muralidhar Kulkarni

1AT22EC 048

Sanika B G

1AT22EC 085

Under the Guidance of

Dr. Prasuna VNP

Associate Professor

Department of ECE



ATRIA
INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
BENGALURU

2025-2026



ATRIA
INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
BENGALURU

Department of Electronics Communication Engineering

CERTIFICATE

Certified that the project work entitled “**IoT Enabled Dynamic Wireless Charging of Electric Vehicle Using Solar Integration**” carried out by

BHAVANI V R

1AT22EC013

KAVANA T R

1AT22EC039

MAHIMA MURALIDHAR KULKARNI

1AT22EC048

SANIKA B G

1AT22EC085

A bonafide students of VII Semester Electronics & Communication Engineering, Atria Institute of Technology, **Bengaluru** in partial fulfillment for the award of Bachelor of Engineering in Visvesvaraya Technological University, Belagavi during the year **2025-26**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Project Guide
Dr.Prasuna VNP
Associate Professor
Atria IT

Head of the Dept
Dr. Keshavamurthy
Professor
Atria IT

Principal
Dr. Rajesha S
Atria IT

Name of the Examiners with Signature and Date:

1.

2.

DECLARATION

We **Ms. Bhavani V.R [1AT22EC013]**, **Ms. Kavana T.R [1AT22EC039]**, **Ms. Mahima Muralidhar Kulkarni [1AT22EC048]** and **Ms. Sanika B.G [1AT22EC085]** students of final year Bachelor of Engineering , Department of Electronics and Communication Engineering, Atria Institute of Technology Bangalore, would hereby declare the project entitled “**IoT Enabled Dynamic Wireless Charging for Electric Vehicle Using Solar Integration**” has been carried out by us at Atria Institute of Technology Bengaluru and submitted in partial fulfillment of the course requirements for the award of degree of Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2025 – 2026.

Place: Bengaluru

Date:

Bhavani V.R

1AT22EC013

Dept. of ECE

Atria IT, Bangalore-24

Kavana T.R

1AT22EC039

Dept. of ECE

Atria IT, Bangalore-24

Mahima Muralidhar

Kulkarni

1AT22EC048

Dept. of ECE

Atria IT, Bangalore-24

Sanika B.G

1AT22EC085

Dept. of ECE

Atria IT, Bangalore-24

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without mention of the people who made it possible, whose constant guidance and encourages crowned my effort of success. We take this opportunity to express deepest gratitude and appreciation to all those who helped us directly or indirectly towards the successful completion of final year project work.

We express my gratitude and respect to **Dr. Rajesha S**, Principal, Atria IT, Bangalore for granting permission to carry out the project.

We express my heartfelt gratitude to **Dr. Keshavamurthy**, Professor and Head of Electronics and Communication Engineering Department, Atria IT, Bangalore for being supportive and encouraging in completing the project.

We express our sincere gratitude to the **Dr. Rakhi S and Prof. Swathi Rathore**, Project co-coordinators, of Electronics and Communication Department, Atria Institute of Technology, for the support extended to us.

We express my deep sense of gratitude to our project guide **Dr. Prasuna VNP** , Associate Professor, Dept. of Electronics and Communication Engineering, Atria IT, Bangalore for motivating and guiding us in successfully carrying out our project.

Finally, we also thank the entire staff of the Dept. of Electronics and Communication Engineering, friends and family for their valuable suggestions and encouragement in all means.

ABSTRACT

Electric Vehicles (EVs) offer a sustainable solution to reduce pollution, fuel dependency, and carbon emissions, yet challenges like slow charging, limited driving range, and grid load hinder their adoption. This project introduces an IoT-enabled Dynamic Wireless Power Transfer (DWPT) system that allows EVs to charge while moving, integrating solar energy, high-frequency inductive power transfer, and smart IoT control to create an efficient and future-ready charging ecosystem. The system incorporates a transformer, rectifier, inverter, amplifier, and transmitter–receiver coils to enable wireless charging, while MPPT ensures optimal use of solar energy and reduced grid dependency. Real-time monitoring, automation, and payment management are achieved through a NodeMCU IoT module, and a 4V battery prototype demonstrates stable charging with 70–80% efficiency at small air gaps. Combining wireless power transfer, IoT intelligence, and solar integration, this project presents a scalable foundation for smart electric roads where EVs can charge seamlessly, enhancing driving range, reducing downtime, and supporting sustainable transportation infrastructure.

TABLE OF CONTENTS

SL NO.		PAGE NO.
1	INTRODUCTION	1
	1.1 Problem statement	3
	1.2 Scope	3
	1.3 Objectives	3
2	LITERATURE SURVEY	4
	Literature Review	5
3	HARDWARE AND SOFTWARE USED	10
	3.1 Hardware Requirements	11
	3.2 Software Used	18
4	METHODOLOGY	21
	4.1 Block Diagram	22
	4.2 Implementation	24
5	RESULT	26
6	APPLICATION,ADVANTAGES AND DISADVANTAGES	31
	6.1 Advantages	32
	6.2 Applications	32
	6.3 Disadvantages	32
7	CONCLUSION AND FUTURE SCOPE	33
	7.1 Conclusion	34
	7.2 Future scope	35
	REFERENCE	

TABLE OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1.1	BRIDGE RECTIFIER	12
3.1.2	STEP DOWN TRANSFORMER	12
3.1.3	PCB BOARD	13
3.1.4	BIPOLAR NPN TRANSISTOR	13
3.1.5	LCD DISPLAY 16X2	14
3.1.6	IoT MODULE NODE MCU	14
3.1.7	IR SENSOR	15
3.1.8	SOLAR PANEL	15
3.1.9	COPPER COIL	16
3.1.10	OLED	16
3.1.11	BATTERY	17
3.1.12	SERVO MOTOR	17
3.1.13	BATTERY MANAGEMENT SYSTEM	18
3.2.1	ARDUINO WINDOW	20
4.1.1	BLOCK DIAGRAM	22
4.2.1	FLOW CHART	24
5.1	POWER SOURCES	27
5.2	TRANSISTOR SECTION	27
5.3	WIRELESS POWER TRANSFER	28
5.4	MICROCONTROLLER AND CONTROL COMPONENTS	28
5.5	DISPLAY OF BATTERY CHARGING	29
5.6	FINAL MODEL	29